

Microscopic events

$$\lambda_{ij}^N = N^{-1} a_i W_{ij} \Psi_{ij}$$

*signed directed events
one event per pair*

 **(C, H) state update**

$$C_j^+ + = \eta R, \quad H_j^+ + = 1$$

$$\text{decay } dC_j = -\delta C_j dt \\ dH_j = -\rho H_j dt$$

**Kinetic limit**

$$\partial_t f = D(f) + \mathcal{I}(f)$$

A_t sender field
type weights π

Finite-time error ($B_i \equiv 1$: no entry/exit)

$$\sup_{0 \leq t \leq T} \mathbb{E} W_1(\mu_t^N, f_t) \leq C_T \left\{ \frac{\log(1+N)}{\sqrt{N}} + \|W_N - W\|_\infty + \varepsilon_{\text{history}} \right\}$$

$$N^{-1/2}$$

particle coupling
(Theorem 2)

$$N^{-1/2}$$

capital and attention
marginals

$$\varepsilon_{\text{history}}$$

pair-history exclusion

$$\varepsilon_{\text{history}}(N, T) \leq (\eta + \omega) \Lambda^2 T^2 e^{L_X T} / (2N) = O(N^{-1}) \quad (\Lambda = AP_1 \max\{1, P_T\}; T^2: \text{accumulated history})$$